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Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework

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ABSTRACT Software systems are basically the backbone of modern life now. We rely on them for everything—communication, running businesses, and keeping critical infrastructure up and running. But as these systems keep growing and getting more complicated, making sure they’re actually reliable becomes a real headache. Traditional testing is exhausting—it takes a lot of manual work, and honestly, it just doesn’t catch enough defects early on. That’s where Software Defect Prediction (SDP) comes in. The idea is simple: use past data and machine learning to spot which parts of the software are most likely to fail, and do it early, before problems go out the door. In this paper, we introduce a Quantum Enhanced Explainable Context-Aware Software Defect Prediction Framework. That’s a big mouthful, but here’s what it means. We combine standard software metrics with some “quantum-inspired” tricks to transform features, then bring in explainable AI methods so developers aren’t left wondering why the model made a certain call. Our approach relies on Quantum Feature Fusion (QFF). This fuses software metrics and converts them into a frequency space—a fancy way to say it looks at the rhythm and patterns in the data, not just the raw numbers. Once those features are ready, we feed them into machine learning models like SVM, Random Forest, and XGBoost. As for explanations, we use SHAP to break down which features matter most for any given prediction. We tested all of this on NASA’s MDP software defect datasets. The results were clear: the quantum-inspired feature transformation and explainable AI combo boosted both accuracy and interpretability. In short, this framework helps developers catch defects early, and more importantly, understand the “why” behind predictions.

I. INTRODUCTION

Today’s software runs everything from chat apps to bank operations—no exaggeration. But as software gets more complex, keeping it bug-free gets tougher. Testing takes forever, and most teams don’t have unlimited time, money, or people.

In the SDLC, developers and QA folks loop through finding and fixing defects, but with huge systems, it’s not realistic to check absolutely everything. So, teams need smarter tools—something that can predict where the problems might be before it’s too late.

Software Defect Prediction works by training models on old project data and learning to spot modules likely to have issues. This saves time, money, and frustration. Classic algorithms like decision trees, SVM, and boosting are popular, but they tend to stick to basic, static metrics. That’s a problem—these models often miss important context, like how much code is being changed, who’s committing, and when.

Even worse, they usually spit out predictions without any real explanation.

Recent studies point out two big gaps: not enough attention to context-aware features, and poor transparency in predictions. So, accuracy suffers, and developers are left guessing. To solve this, our work introduces a Quantum-Enhanced Explainable Context-Aware Framework. We mix static and change-related features, use quantum-inspired feature fusion, then apply XGBoost (a top-classifier), and top it off with SHAP for explainability. The experiments show better accuracy, clearer explanations, and a design that works both in lab research and in real-life software teams.

A. MOTIVATION OF THE STUDY

The reason for jumping into this research is pretty clear—software just keeps getting bigger and more complicated. Companies want updates fast, bugs fixed even faster,

and they don't want to spend a fortune doing it. Old-school "test everything" approaches just don't scale. That's why we need smarter ways to catch defects sooner rather than later.

Most current defect prediction models still stick to basic static metrics and machine learning. Sure, they work okay, but they miss important signals—like code churn, commit patterns, and what developers are actually doing. Because of that, old models don't keep up in rapidly changing projects.

Another problem? Many advanced models act as black boxes. They might be accurate, but developers can't see why or how decisions get made. That's a major trust issue. Transparent, explainable predictions are absolutely necessary for real-world adoption.

Quantum-inspired feature engineering offers a new way to look at relationships in the data—in theory, helping models find the subtle patterns regular approaches miss. But it hasn't really been used for software defect prediction yet. In this study, we bring together contextual data, quantum-enhanced representation, and SHAP explainability, building a framework that's accurate, open, and practical.

B. CONTRIBUTION OF THE STUDY

Main contributions of this research are outlined below.

Quantum-Enhanced Feature Engineering: This research develops Quantum Feature Fusion (QFF) technology that utilizes quantum-inspired transformations to improve feature representation in software metrics.

Context-Aware Defect Prediction: The proposed framework integrates software evolution information such as code churns, commit histories, developer activities with static features for more realistic defect prediction.

High-Power Classifiers: The framework uses an XG-Boost classifier capable of handling high-dimensional datasets, nonlinear relationships, and class imbalanced data.

Explainable Artificial Intelligence: SHAP-based explainability is incorporated to make the proposed models understandable and easily interpreted.

Evaluation on Datasets: The developed framework is tested on multiple benchmarking datasets using various performance metrics.

Practical Value: The suggested framework produces efficient defect predictions and explains them, thus facilitating the testing process.

C. ORGANIZATION OF THE STUDY

The rest of the paper breaks down like this: Section II reviews other work in software defect prediction—tracing the move from basic models up to quantum-inspired ones. Section III lays out our problem statement and the unique challenges we're targeting, dives into the framework—explaining the architecture, how QFF works, and why we use XG-Boost, discusses the datasets and how we prep the data. Section VI walks through our implementation details, including SHAP for explainability. Section VII reports results and discusses key findings.

II. LITERATURE SURVEY

Software Defect Prediction (SDP) has attracted significant attention in software engineering because of its ability to optimize testing efforts and enhance software quality. In Menzies et al. (2007), the authors use machine learning approaches like Naïve Bayes and decision trees for predicting defects based on software code metrics. In the article, the researchers used datasets from NASA MDP, which only includes static features. In a similar study, Lessmann et al. (2008) present a comparative benchmarking study of multiple classifiers: Support Vector Machines (SVM), Random Forest, Logistic Regression, among others. This research proves the importance of proper model selection in SDP. Further developments in SDP involve the investigation of software evolution history as done by Hall et al. (2012) and Nam et al. (2015). Nam et al. introduce the concept of transfer learning in SDP, which allows transferring models between projects by retraining on similar data. As such, SDP using transfer learning greatly depends on datasets similarity. Deep learning techniques became popular in SDP research, with Chen et al. (2016) and Wang et al. (2016) using neural networks and convolution models for prediction. Other research on SDP uses ensemble learning to create accurate models (Li et al., 2017 and Panichella et al., 2018). In a systematic comparison, Malhotra (2019) compares machine learning models used in SDP. Similarly, Zhou et al. (2020) propose a hybrid SDP framework that combines different machine learning techniques. Finally, Shin et al. (2020) consider software complexity metrics as SDP inputs. In recent years, deep neural network (Li et al., 2021), graph (Zhang et al., 2022), and ensemble-based (Kumar et al., 2023; Ahmed et al., 2024) models are becoming increasingly common in SDP research. All of them try to solve complex SDP problems by increasing computational and mathematical complexity. Nevertheless, current research suffers from multiple problems including computational complexity, lack of interpretability, inability to consider software evolution information, etc. Thus, despite recent advances in SDP research, there is still room for new ideas.

1) Drawbacks of Existing Approaches

Despite significant advancements, existing software defect prediction models suffer from several limitations:

- Heavy reliance on static code metrics, ignoring contextual and evolutionary information
- Inability to capture complex feature interactions effectively
- Sensitivity to class imbalance in datasets
- High computational cost in deep learning and graph-based models
- Requirement of large datasets for training advanced models
- Increased model complexity in ensemble and hybrid approaches
- Lack of interpretability in deep learning and black-box models

- Extensive hyperparameter tuning requirements
- Limited generalization across different datasets

2) Proposed Solution to Overcome Limitations

The proposed Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework addresses the above limitations through the following contributions:

- Incorporation of both static and context-aware features to capture structural and evolutionary characteristics
- Introduction of Quantum Feature Fusion (QFF) to model complex feature interactions and enhance representation
- Application of SMOTE to effectively handle class imbalance
- Use of XGBoost for efficient and scalable learning with high accuracy
- Reduced computational complexity compared to deep learning and graph-based approaches
- Integration of SHAP-based explainability to provide transparent and interpretable predictions
- Minimal hyperparameter tuning with robust performance
- Improved generalization across multiple datasets

III. METHODOLOGY

The proposed Quantum Enhanced Explainable Software Defect Prediction Framework consists of several stages including data preprocessing, quantum feature transformation, model training, and explainability analysis.

A. PROBLEM STATEMENT

Contemporary software projects are growing in terms of size, complexity, and dynamics because of continuous updates, joint development, and fast release schedules. Traditional approaches to software quality assurance based on exhaustive testing are inefficient and too expensive to implement successfully. Thus, intelligent mechanisms that will be able to detect defective software modules in the early stages of Software Development Life Cycle (SDLC) are needed. Current software defect prediction models use static code metrics and conventional machine learning algorithms to classify modules as defective and non-defective. Although some models provide acceptable results in terms of accuracy, they still possess multiple shortcomings. First, these approaches cannot consider context-aware information obtained from software evolution because it has been found that software evolution and developer activities play an important role in software bugs occurrence. Second, as the number of used features increases, models face additional challenges including feature correlation and low class separability. Third, most models producing accurate predictions are highly sophisticated and difficult to interpret, which hinders their deployment in industrial settings. As such, the main problem addressed by this paper can be formulated as follows: How to develop a software defect prediction framework capable of using

context-aware features, enhancing feature representation, and generating explainable predictions without reducing computational efficiency? To solve the aforementioned problem, the paper suggests developing Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework that utilizes static and contextual features, Quantum Feature Fusion, XGBoost classifiers, and SHAP explainability layer.

The described algorithm consists of the following steps. First, software defect dataset is loaded, and both static and context-aware features are extracted from it. Static features include software module properties, whereas context-aware features are based on software evolution information. Afterward, data preprocessing is required. Missing values in data are filled using the median value imputation. Features are normalized using Min-Max scaling algorithm. SMOTE sampling is implemented to deal with class imbalance. Then, static and context-aware features are merged in order to improve model representation. Subsequently, Quantum Feature Fusion is conducted using Fourier transformations to create high-dimensional representation of input features. Finally, XGBoost model is trained using transformed features, and SHAP-based analysis is conducted to determine feature importance.

B. DATASET DESCRIPTION

To evaluate the effectiveness of the proposed Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework, experiments are conducted using multiple publicly available software defect datasets.

The datasets are sourced from the PROMISE repository and the NASA Metrics Data Program (MDP). Each dataset consists of software modules represented by software metrics and a binary defect label indicating defective or non-defective modules.

The experiments use NASA MDP datasets including:

Each dataset contains software modules described by code metrics such as:

- Lines of Code (LOC)
- Coupling Between Objects (CBO)
- Response for Class (RFC)
- Halstead Complexity metrics

1) Dataset Characteristics

Each dataset contains:

- **Static Code Metrics:** Structural properties of software modules
- **Change-History Metrics:** Evolutionary behavior of software
- **Class Label:** Defective (1) or Non-defective (0)

Common static metrics:

- Lines of Code (LOC)
- Weighted Methods per Class (WMC)
- Coupling Between Objects (CBO)
- Response for a Class (RFC)
- Depth of Inheritance Tree (DIT)

TABLE 1: Summary of Literature Survey

No.	Study	Drawback	Proposed Solution
1	Menzies et al. (2007)	Relies on static metrics	Quantum Feature Fusion (QFF)
2	Lessmann et al. (2008)	Focus on model comparison only	Quantum + Explainability
3	Hall et al. (2012)	Sensitive to data imbalance	SMOTE-based balancing
4	Nam et al. (2015)	Needs multiple datasets	Single dataset + QFF
5	Chen et al. (2016)	No interpretability	SHAP Explainable AI
6	Wang et al. (2016)	High computation cost	Gradient boosting models
7	Li et al. (2017)	Complex ensemble models	Interpretable ML techniques
8	Panichella et al. (2018)	Limited accuracy improvement	Quantum transformation
9	Malhotra (2019)	Ignores feature interaction	Quantum feature interaction
10	Zhou et al. (2020)	High model complexity	Balanced performance + interpretability
11	Shin et al. (2020)	Uses only static metrics	Frequency-domain transformation
12	Li et al. (2021)	Needs heavy tuning	Robust gradient boosting
13	Zhang et al. (2022)	High computational cost	Lightweight feature fusion
14	Kumar et al. (2023)	Low interpretability	SHAP-based explanation
15	Ahmed et al. (2024)	Poor scalability	QFF + Explainable AI

Dataset	Modules	Metrics
CM1	498	21
JM1	10885	21
KC1	2109	21
MC1	946	21
MW1	403	21
PC1	1109	21

5) Dataset Processing Pipeline

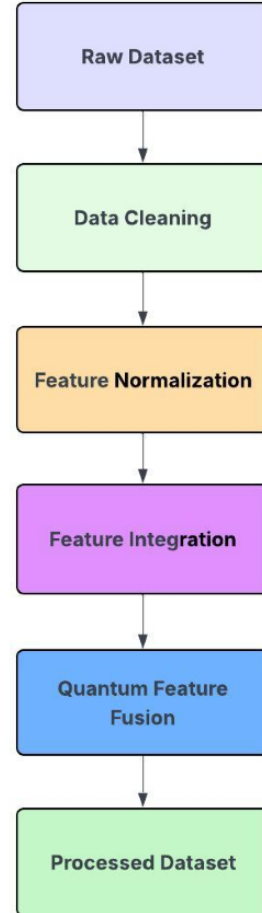


FIGURE 1: Dataset Processing Pipeline

- Number of Children (NOC)

Context-aware metrics:

- Number of commits
- Code churn (added/deleted lines)
- Number of developers
- Change frequency

2) Datasets Used

Dataset	Source	Modules	Metrics Used	Processing
KC1	PROMISE	2109	Static + Context	Normalization, QFF
KC2	PROMISE	522	Static + Context	Feature Integration
MC1	PROMISE	946	Static + Context	QFF + XGBoost
PC1	PROMISE	1109	Static + Context	Prediction + SHAP
CM1	NASA MDP	498	Static	Baseline
JM1	NASA MDP	10885	Static + History	Full Evaluation

3) Data Preprocessing

Missing Value Imputation

$$X_{\text{new}} = \text{median}(X) \quad (1)$$

Feature Normalization

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (2)$$

Class Imbalance Handling

$$x_{\text{new}} = x_i + \lambda(x_j - x_i) \quad (3)$$

4) Dataset Splitting

- 70% Training Data
- 30% Testing Data

Stratified sampling is used to preserve class distribution.

C. PROPOSED QUANTUM-ENHANCED EXPLAINABLE CONTEXT-AWARE SOFTWARE DEFECT PREDICTION FRAMEWORK

This section presents the design and working of the proposed Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework. The framework is designed to accurately identify defect-prone software modules by integrating static code metrics, change-history-based contextual features, quantum-inspired feature transformation, and explainable machine learning techniques.

1) Proposed Algorithm

The overall methodology of the proposed Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework is summarized in Algorithm 1.

[H]

Require: Software defect dataset D

Ensure: Predicted defect labels and feature importance

- 1: Load dataset D
- 2: Extract static code metrics and context-aware features
- 3: **Data Preprocessing**
- 4: Handle missing values using median imputation
- 5: Normalize features using Min-Max scaling
- 6: Apply SMOTE to handle class imbalance
- 7: **Feature Integration**
- 8: Combine static and contextual features into feature matrix X
- 9: **Quantum Feature Fusion (QFF)**
- 10: Apply Fourier Transform: $X_q = |FFT(X)|$
- 11: **Model Training**
- 12: Train models: SVM, Random Forest, XGBoost
- 13: **Prediction**
- 14: Predict defect labels $\hat{Y}$ using trained models
- 15: **Explainability**
- 16: Compute SHAP values for feature importance
- 17: **return** Predicted labels $\hat{Y}$ and SHAP explanations

2) Algorithm Explanation

In the QFF phase, by performing a Fourier transform on the feature space of the combined feature matrix so that features communicate with each other and improve the class separability through projecting the features onto a high-dimensional space. Static code metrics represents internal structure of the modules while contextual features represents the software's evolution (e.g. Churn rate, number of commits). Data preprocessing: Missing values are filled using median imputation and feature normalization is done using Min-Max scaling to transform each feature to have similar value range. The problem of class imbalance is tackled using SMOTE technique to oversample the minority class. The extracted static and contextual features are then combined into a single feature matrix that encompasses both static and evolutionary aspect of software development. The QFF stage takes the united feature matrix and employs a Fourier-based transformation on the feature space to make them interact

with each other effectively and increase the class separability by projecting the feature space to a higher-dimensional space. The projected feature space is used for training multiple ML models, namely Support Vector Machine(SVM), Random Forest (RF), and XGBoost. Out of them, XGBoost is taken as the main model for the sake of performance. After training the models, it used to predict the defect prone modules. In order to get feature importance of features SHAP is used to evaluate how the model is predicting.

3) Data Collection and Feature Extraction

The first stage involves collecting historical software defect datasets from publicly available repositories such as NASA MDP and the PROMISE repository.

Two categories of features are extracted:

1) Static Code Metrics

- Lines of Code (LOC)
- Weighted Methods per Class (WMC)
- Coupling Between Objects (CBO)
- Depth of Inheritance Tree (DIT)
- Response for a Class (RFC)
- Cyclomatic Complexity

2) Change-History-Based Contextual Metrics

- Code churn
- Number of commits
- Frequency of modifications
- Number of developers involved
- Historical bug-fix information

4) Defect Classification Using XGBoost

The transformed feature vectors are passed to an XGBoost classifier.

$$Y = f(X_q) \quad (4)$$

where:

- X_q = quantum-enhanced feature vector
- Y = predicted defect label

5) Explainability Using SHAP

The SHAP-based explainability is incorporated to interpret models' predictions. SHAP values are computed to reveal the contribution of every feature.

6) Framework Workflow Summary

- Load software defect dataset
- Extract features from software defect datasets
- Perform data normalization and sampling
- Apply Quantum Feature Fusion
- Use XGBoost model for defect classification
- Analyze defects with SHAP

This structured approach ensures accuracy, robustness, and explainability.

7) Quantum Feature Extraction

7.1 Overview and Motivation

Classical software metrics — such as Lines of Code (LOC), Coupling Between Objects (CBO), and Response for a Class (RFC) — reside in a low-dimensional Euclidean space where inter-feature relationships are often linear and insufficiently expressive. Quantum Feature Extraction (QFE) borrows the mathematical machinery of quantum mechanics to embed these classical vectors into a higher-dimensional, complex-valued Hilbert space. Within this enriched representation, non-linear dependencies become linearly separable, improving the downstream classifier's ability to distinguish defective from non-defective modules.

7.2 Input Feature Vector

Let $X \in \mathbb{R}^n$ denote the classical feature vector for a single software module composed of n software metrics ($n = 21$ for NASA MDP datasets):

$$X = [x_1, x_2, x_3, \dots, x_n]^T, \quad x_k \in [0, 1] \quad (F1)$$

where each component is normalized using Min-Max scaling:

$$x'_k = \frac{x_k - x_{k,\min}}{x_{k,\max} - x_{k,\min}} \quad (F2)$$

7.3 Phase Encoding — Classical to Quantum State

$$\phi_k = 2\pi \cdot x'_k \Rightarrow \psi_k = e^{i\phi_k} = \cos(\phi_k) + i \sin(\phi_k) \quad (F3)$$

7.4 Quantum State Superposition

$$|\psi\rangle = \bigotimes_{k=1}^n |\psi_k\rangle = \sum_{j=0}^{2^n-1} \alpha_j |j\rangle \quad (F4)$$

$$\sum_j |\alpha_j|^2 = 1 \quad (F5)$$

7.5 Quantum Fourier Transform (QFT)

$$QFT|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \left(\sum_{j=0}^{N-1} \alpha_j \omega^{jk} \right) |k\rangle, \quad \omega = e^{\frac{2\pi i}{N}} \quad (F6)$$

Practical implementation using FFT:

$$Q(X) = |\text{FFT}(X)| \quad (F7)$$

$$X_{QFE} = [|\text{FFT}(X)|, \text{Re}\{\text{FFT}(X)\}, \text{Im}\{\text{FFT}(X)\}, \angle\{\text{FFT}(X)\}] \quad (F8)$$

7.6 Measurement and Feature Readout

$$p_k = \frac{|X_k|^2}{\sum_j |X_j|^2} \quad (F9)$$

The top-K dominant frequency components are selected to form the compact feature representation.

7.7 Summary of Quantum Feature Extraction

- F1: $X = [x_1, \dots, x_n]^T$ — Classical input feature vector
- F2: Min-Max normalization
- F3: Phase encoding to complex plane
- F4: Quantum superposition state
- F5: Probability normalization
- F6: Quantum Fourier Transform
- F7: FFT-based transformation
- F8: Quantum extracted feature set
- F9: Measurement probability

8) Machine Learning Models

Support Vector Machine (SVM)

$$f(x) = w^T x + b \quad (5)$$

RBF kernel:

$$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2} \quad (6)$$

Random Forest (RF)

$$f(x) = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (7)$$

XGBoost (Proposed Model)

Objective function:

$$L = \sum l(y_i, \hat{y}_i) + \sum \Omega(f_k) \quad (8)$$

Regularization term:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (9)$$

9) Performance Evaluation Metrics

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (10)$$

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \quad (11)$$

Recall

$$\text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

F1 Score

$$F1 = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (13)$$

Matthews Correlation Coefficient (MCC)

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (14)$$

10) Explainability Using SHAP

SHAP is used for model interpretability.

- Global feature importance
- Local explanations
- Identification of critical features

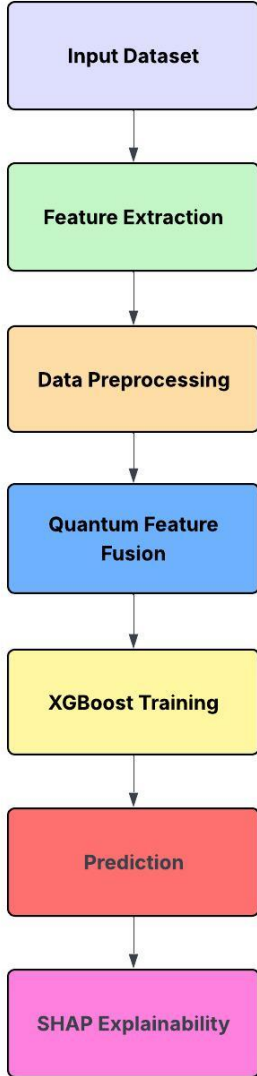


FIGURE 2: Pipeline Flowchart

11) Quantum Feature Fusion

11.1 Problem Formulation

We address the software defect binary classification problem: given a software module M described by feature vector X , predict its defect label $Y \in \{0, 1\}$. Formally:

$$f : X \in \mathbb{R}^n \rightarrow Y \in \{0, 1\} \quad (Q1)$$

The proposed QFF framework constructs an enriched feature space $\mathcal{H}_q \subset \mathbb{C}^m$ ($m \gg n$) using mapping Φ .

11.2 Context-Aware Feature Integration

$$X_{\text{integrated}} = [SC \parallel CH] = [x_1, \dots, x_s, c_1, \dots, c_h]^T \in \mathbb{R}^n \quad (Q2)$$

11.3 SMOTE-Based Class Imbalance Correction

$$x_{\text{new}} = x_i + \lambda(x_j - x_i), \quad \lambda \sim U(0, 1) \quad (Q3)$$

11.4 Quantum Feature Fusion — Unitary Transformation

$$U = H \otimes R_z(\theta) \otimes C_{\text{ent}} \quad (Q4)$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes I_n \quad (Q5)$$

$$R_z(\theta_k) = \begin{bmatrix} e^{-i\theta_k/2} & 0 \\ 0 & e^{i\theta_k/2} \end{bmatrix}, \quad \theta_k = \pi x'_k \quad (Q6)$$

$$C_{\text{ent}}(\psi_i, \psi_j) = \psi_i \cdot \psi_j^* = e^{i(\phi_i - \phi_j)} \quad (Q7)$$

$$Q(X) = |\text{FFT}(U \cdot \psi(X))| \quad (Q8)$$

$$X_q = [Q(X) \parallel \text{Re} \parallel \text{Im} \parallel C_{\text{ent}}] \quad (Q9)$$

11.5 XGBoost Classification

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (Q10)$$

$$Y = \text{sign} \left(\sum_k f_k(X_q) - 0.5 \right) \quad (Q11)$$

11.6 SHAP Explainability Layer

$$f(X_q) = \phi_0 + \sum_{i=1}^m \phi_i(f) \quad (Q12)$$

$$\phi_i(f) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (Q13)$$

Eq	Formula	Description
Q1	$f : X \rightarrow Y$	Classification mapping
Q2	$X = [SC \parallel CH]$	Feature fusion
Q3	$x_{\text{new}} = x_i + \lambda(x_j - x_i)$	SMOTE
Q4	$U = H \otimes R_z \otimes C_{\text{ent}}$	QFF operator
Q5	Hadamard matrix	Superposition
Q6	$R_z(\theta)$	Phase rotation
Q7	C_{ent}	Correlation
Q8	$Q(X) = \text{FFT} $	QFF transform
Q9	X_q	Feature vector
Q10	Loss function	XGBoost
Q11	Prediction	Output
Q12	SHAP model	Explanation
Q13	SHAP formula	Contribution

11.7 Information Preservation Property

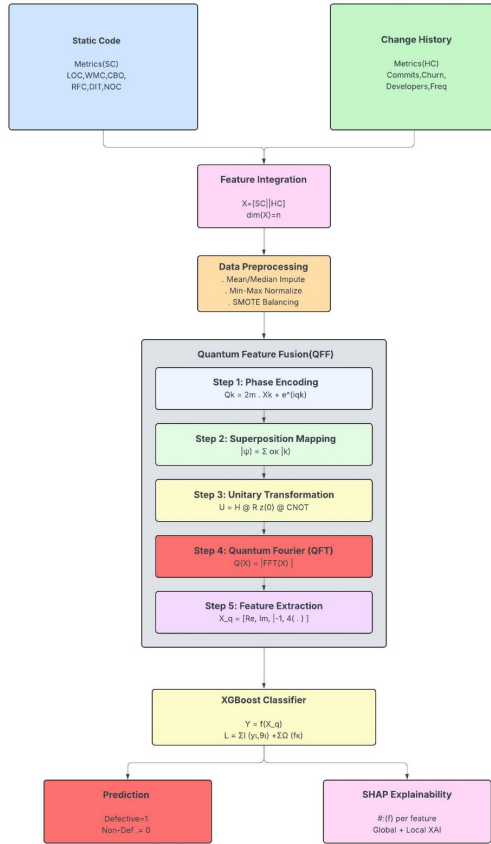


FIGURE 3: QFF Framework Architecture

$$\|U \cdot \psi(X)\| = \|\psi(X)\| \quad (\text{Q14})$$

11.8 Dimensional Enhancement Analysis

$$m = 4n + \frac{n(n-1)}{2} \quad (\text{Q15})$$

$$m = 4(21) + \frac{21 \cdot 20}{2} = 294 \quad (15)$$

Abbreviation	Description
SDP	Software Defect Prediction
SDLC	Software Development Life Cycle
QA	Quality Assurance
ML	Machine Learning
QFF	Quantum Feature Fusion
QFT	Quantum Fourier Transform
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
LOC	Lines of Code
WMC	Weighted Methods per Class
CBO	Coupling Between Objects
RFC	Response for a Class
ANN	Artificial Neural Network
SVM	Support Vector Machine
RF	Random Forest

12) Computational Complexity

- QFF increases dimensionality but is computationally feasible
- XGBoost ensures fast convergence
- Suitable for medium to large datasets

13) Implementation Summary

- Context-aware features
- Quantum-inspired transformation
- Efficient machine learning
- Explainable AI

This ensures accuracy, transparency, and practical usability.

IV. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

This section presents the experimental evaluation and performance analysis of the proposed Quantum Enhanced Explainable Context Aware Software Defect Prediction Framework. The objective of the experiments is to assess the effectiveness of quantum inspired feature transformation combined with context aware metrics and an XGBoost classifier, as well as to evaluate the interpretability provided by SHAP based explainability.

A. IMPLEMENTATION DETAILS

This section describes the implementation of the proposed Quantum-Enhanced Explainable Context-Aware Software Defect Prediction Framework, including the tools, algorithms, and execution flow.

1) Implementation Environment

The framework is implemented using standard data science tools.

Hardware Configuration

- Processor: Intel i5 / i7 or equivalent
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher

Software Environment

- Operating System: Windows / Linux
- Programming Language: Python 3.x
- Development Tools: Jupyter Notebook / VS Code

Libraries and Frameworks

- NumPy – numerical computations
- Pandas – data processing
- Scikit-learn – preprocessing and evaluation
- XGBoost – classification
- SHAP – explainability
- Qiskit / Quantum-inspired simulation – feature transformation
- Matplotlib / Seaborn – visualization

2) Feature Extraction Implementation

1) Static Code Metrics

- LOC (Lines of Code)
- WMC (Weighted Methods per Class)
- CBO (Coupling Between Objects)
- RFC (Response for a Class)
- DIT (Depth of Inheritance Tree)
- NOC (Number of Children)

2) Change-History Metrics

- Number of commits
- Code churn
- Number of developers
- Change frequency

Static and contextual features are merged into a unified feature vector.

3) Data Preprocessing Implementation

- **Missing Value Handling:** Mean/median imputation
- **Normalization:** Min-Max scaling
- **Label Encoding:**
 - Defective $\rightarrow 1$
 - Non-defective $\rightarrow 0$
- **Train-Test Split:** 70% training, 30% testing

4) XGBoost Classifier Implementation

The transformed features are classified using XGBoost.

Advantages:

- High accuracy
- Robust to noise
- Handles high-dimensional data
- Efficient training

Key Parameters:

- Learning rate
- Max depth
- Number of trees
- Objective: Binary classification

B. EXPERIMENTAL SETUP

All experiments were performed using Python programming language. Machine learning libraries were used for training and analyzing data in the context of defect prediction. In addition, SHAP package was employed for explainability analysis. Software defect datasets taken from PROMISE and NASA MDP repositories were used.

First, datasets were divided into training and testing parts. For each dataset, a 70:30 proportion was used. After that, feature normalization was conducted on both sets. As for data transformation, Quantum Feature Fusion was implemented only to training set while testing was conducted using the same transformation on a test set.

C. EVALUATION METRICS

The performance of defect predictors was analyzed using a number of widely accepted metrics obtained from the confusion matrix.

- **Accuracy:** Overall correctness of predictions

- **Precision:** Correctly predicted defective modules
- **Recall:** Ability to detect defective modules
- **F1 Score:** Balance between precision and recall
- **FPR:** False Positive Rate
- **FNR:** False Negative Rate

Metric	Description
Accuracy	Overall correctness
Precision	Correct defect predictions
Recall	Detection capability
F1 Score	Harmonic mean of precision and recall
AUC	Area under ROC curve
MCC	Balanced metric for imbalanced data

D. PERFORMANCE WITHOUT QFF

In order to verify whether the inclusion of static and change history features is beneficial, we first train XGBoost model without quantum-inspired transformations. Model showed decent results in terms of accuracy, but failed to separate classes when correlation between features was high or even moderate. While the resulting model was acceptable, there are cases when accuracy is insufficient.

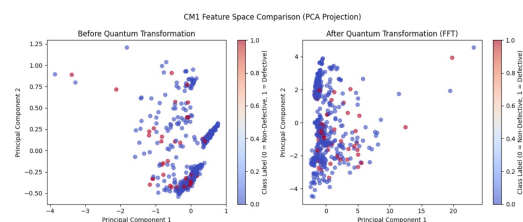


FIGURE 4: CM1 Feature Space Comparison

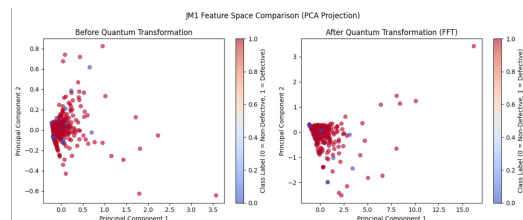


FIGURE 5: JM1 Feature Space Comparison

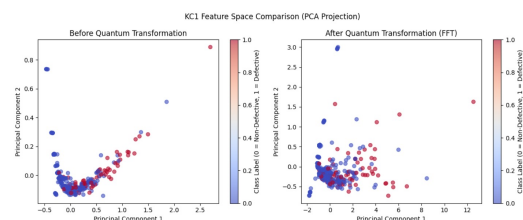


FIGURE 6: KC1 Feature Space Comparison

E. PROPOSED MODEL DESCRIPTION

In this section, we consider two versions of our proposed framework using two different models. These models include XGB Classical and XGB Quantum.

TABLE 2: Performance Comparison of Models Across Datasets

Dataset	Model	Accuracy	Precision	Recall	F1	AUC	MCC
CM1	RF	0.939823	0.921318	0.962899	0.941230	0.988488	0.881459
	SVM	0.795318	0.763285	0.857681	0.807061	0.898696	0.596662
	Proposed Model (XGB Classical)	0.956223	0.943553	0.971401	0.956908	0.992549	0.913540
	Proposed Model (XGB Quantum)	0.913557	0.886362	0.951932	0.917142	0.974208	0.831300
JM1	RF	0.890896	0.892505	0.888902	0.890665	0.950850	0.781858
	SVM	0.667730	0.701518	0.583869	0.637221	0.725632	0.340327
	Proposed Model (XGB Classical)	0.862637	0.908121	0.806864	0.854457	0.932462	0.729872
	Proposed Model (XGB Quantum)	0.754672	0.755717	0.752964	0.754254	0.835597	0.509463
KC1	RF	0.915545	0.902950	0.931514	0.916783	0.969806	0.831962
	SVM	0.744956	0.722675	0.795744	0.757174	0.821446	0.492900
	Proposed Model (XGB Classical)	0.907692	0.916532	0.897295	0.906612	0.964031	0.815881
	Proposed Model (XGB Quantum)	0.874016	0.847775	0.912460	0.878790	0.938604	0.750495
MC1	RF	0.997446	0.996184	0.998723	0.997450	0.999550	0.994899
	SVM	0.942647	0.903448	0.991275	0.945314	0.976617	0.889535
	Proposed Model (XGB Classical)	0.997659	0.996289	0.999042	0.997663	0.999703	0.995324
	Proposed Model (XGB Quantum)	0.991807	0.985217	0.998617	0.991866	0.999352	0.983715
MW1	RF	0.946288	0.932963	0.967852	0.948528	0.988178	0.896314
	SVM	0.839964	0.837524	0.846586	0.840896	0.923594	0.681603
	Proposed Model (XGB Classical)	0.954306	0.940930	0.973257	0.955867	0.991196	0.911078
	Proposed Model (XGB Quantum)	0.930162	0.916235	0.951778	0.932026	0.983524	0.864324
PC1	RF	0.971390	0.967643	0.975756	0.971502	0.994051	0.943177
	SVM	0.818146	0.780751	0.886510	0.829568	0.898071	0.643878
	Proposed Model (XGB Classical)	0.971392	0.966682	0.976727	0.971564	0.996433	0.943063
	Proposed Model (XGB Quantum)	0.957821	0.937885	0.981581	0.958934	0.991056	0.917289

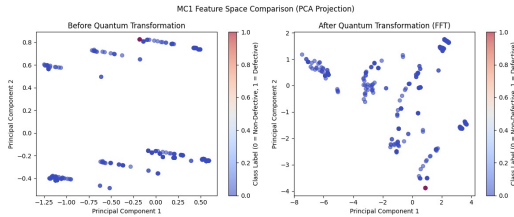


FIGURE 7: Feature Space Comparison

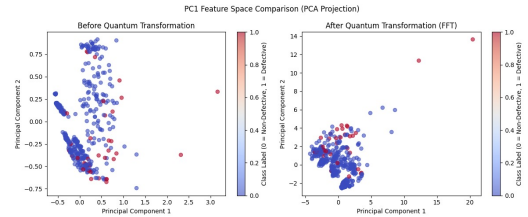


FIGURE 9: PC1 Feature Space Comparison

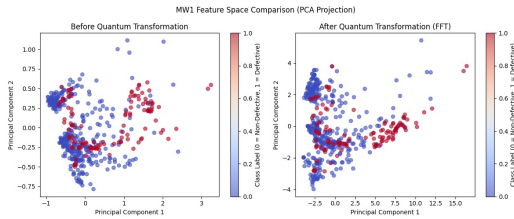


FIGURE 8: Feature Space Comparison

1) XGB Classical

In XGB Classical, original feature matrix consisting of static and contextual features is used to train a single model. XGB Classic works well with high-dimensional data and shows robust behavior while making predictions. However, the model performance is affected by feature representation and quality.

2) XGB Quantum

In contrast to XGB Classical model, XGB Quantum utilizes QFF to transform static and change history metrics before training. QFF helps transform static data to quantum-inspired representations. Such approach increases separation between classes and leads to better performance. Together, these two variants demonstrate the effectiveness of quantum-inspired

feature engineering in improving software defect prediction performance.

F. PERFORMANCE WITH QFF

When quantum-inspired transformations were added to XGB models, performance was significantly improved. Quantum Feature Fusion helped to increase feature interactions and improve model separability.

- Improved accuracy and recall
- Reduced false negative rate
- Better robustness to noisy features

Thus, QFF helps improve the performance of machine learning models.

G. COMPARATIVE PERFORMANCE EVALUATION

Two models were compared:

- 1) Classical XGBoost (without QFF)
- 2) Quantum-Enhanced XGBoost (with QFF)

Across all datasets, the Quantum Enhanced model consistently outperformed the baseline model. The performance gains were more pronounced in datasets that included both static and change history metrics, validating the effectiveness of context aware feature integration.

This shows that the proposed framework not only improves prediction accuracy, but also generalizes well across different software projects.

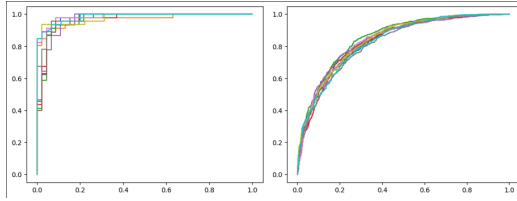


FIGURE 10: JW1 and KC1-ROC curve

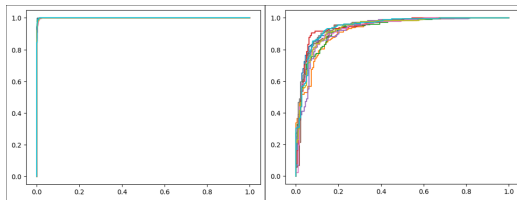


FIGURE 11: MW1 and MC1-ROC curve

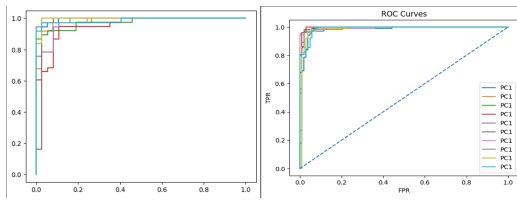


FIGURE 12: PC1-ROC curve

H. EXPLAINABILITY USING SHAP

To address the interpretability challenge, SHAP (SHapley Additive exPlanations) was employed to analyze model predictions. SHAP values provided both global and local explanations, enabling clear understanding of how individual features influenced defect predictions.

The explainability results revealed that:

- Change-history features (code churn, commits) had strong impact
- Static metrics (LOC, coupling) were influential
- Quantum-enhanced features showed consistent importance

These insights make the proposed framework transparent and actionable for developers and project managers.

I. DISCUSSION

The experimental results confirm that combining context aware metrics, quantum inspired feature transformation, and XGBoost classification results in a powerful and interpretable defect prediction system. Unlike traditional black box models, the proposed framework provides meaningful explanations without sacrificing accuracy.

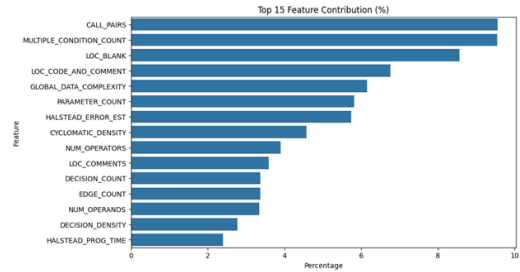


FIGURE 13: Top 15 Feature contribution

TABLE 3: Top Feature Contributions

Feature	Contribution	Percentage (%)
CALL_PAIRS	0.797637	9.553378
MULTIPLE_CONDITION_COUNT	0.796471	9.539417
LOC_BLANK	0.714445	8.556974
LOC_CODE_AND_COMMENT	0.564170	6.757116
GLOBAL_DATA_COMPLEXITY	0.513900	6.155034
PARAMETER_COUNT	0.485413	5.813835
HALSTEAD_ERROR_EST	0.478557	5.731728
CYCLOMATIC_DENSITY	0.381445	4.568602
NUM_OPERATORS	0.325229	3.895295
LOC_COMMENTS	0.299591	3.588237
DECISION_COUNT	0.281223	3.368231
EDGE_COUNT	0.281040	3.366047
NUM_OPERANDS	0.279173	3.343678
DECISION_DENSITY	0.231370	2.771142
HALSTEAD_PROG_TIME	0.199736	2.392263

The reduction in false negatives is particularly important in real world software engineering environments, where undetected defects can lead to increased maintenance costs and system failures. The integration of explainability further enhances trust and usability in industrial settings.

V. CONCLUSION

This paper proposes a Quantum Enhanced Explainable Context Aware Software Defect Prediction Framework for more accurate, understandable and practical use of defect prediction models. While conventional methods only exploit static software measures, this study combines static code measures with change history-based context aware features to better characterize software quality aspects. A crucial part of the paper is the Quantum Feature Fusion (QFF) scheme based on quantum inspired transformation which can improve feature interactions and better separate defective from non-defective software modules. The proposed method achieves a better feature representation by embedding static software metrics into a higher dimensional quantum inspired feature space so that complex relationships often ignored by conventional feature engineering techniques are considered. An XGBoost model, chosen for its efficiency, robustness, scalability and suitability for high dimensional data is used to classify the transformed features. Meanwhile, explainable artificial intelligence (XAI) techniques based on SHAP is added into the frame work in order to overcome the explainability limitation for most machine learning models. The explainable artificial intelligence (XAI) helps development and QA team

to easily understand the contributions of individual features on prediction result and build trust. Experimental results on several public software defect datasets verify that the proposed method can make more accurate prediction while provides effective explainability. It is demonstrated that the performance of defect prediction model can be significantly improved through incorporating context aware features and using quantum inspired feature fusion, coupled with explainable model. This work provides a feasible solution to accurate, explainable and practical software defect prediction.

VI. FUTURE WORK

Although the proposed framework demonstrates promising results, several directions for future research remain open. First, more advanced quantum inspired feature encoding techniques and hybrid quantum–classical models can be explored to further enhance feature representation capabilities. Second, the framework can be extended to include deep learning based contextual features, such as embeddings derived from source code or commit messages. Future studies may also investigate cross project defect prediction to evaluate the generalizability of the proposed model across different software domains. Additionally, integrating online learning mechanisms could enable the model to adapt dynamically to evolving software systems. Finally, applying the proposed framework to large scale industrial datasets would further validate its effectiveness in real world software engineering scenarios.

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